



DAY 2: Networking Basics: IP, MAC, DNS, DHCP, Ports, Protocols · TCP/UDP

IP Address:

An **IP address** is a unique numerical identifier assigned to every device connected to a network. It enables devices to communicate and exchange data over the internet or a local network.

Mac Address:

A **MAC address** is a unique hardware identifier assigned to a network interface card (NIC). It is used to identify devices within a local network.

DNS (Domain Name System)

DNS is the system that translates human-readable domain names (e.g., google.com) into IP addresses that computers use to locate servers.

DHCP (Dynamic Host Configuration Protocol)

DHCP is a network protocol that automatically assigns IP addresses and other network settings to devices, eliminating the need for manual configuration.

Ports

A **port** is a logical communication endpoint on a device that allows different applications and services to send and receive data over a network. For example, HTTP uses port **80**, and HTTPS uses port **443**.

Protocols

A **protocol** is a set of rules that defines how data is transmitted and received between devices on a network. Common protocols include **HTTP**, **HTTPS**, **FTP**, **SSH**, **TCP**, and **UDP**.

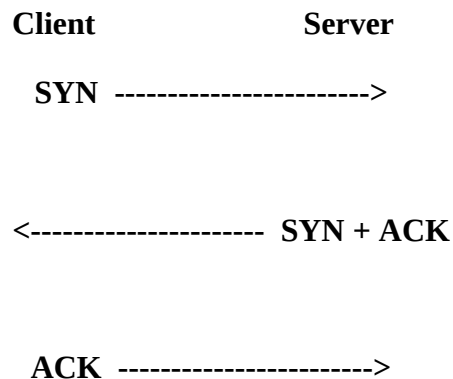
TCP:

TCP stands for Transmission Control Protocol.

It is a connection oriented protocol means it establish a connection between sender and receiver
It ensure data must be receive complete, receive is same order, if a packet loss then resend it and remove the duplicate packets

TCP 3-Way Handshake:

when client connect to the server then there is three steps



Step 1:

Client send a packet to the server *ACK*.

Step 2:

Server Replies *SYN + ACK* means server replies connection request receive and i am ready to communicate

Step 3:

Client sent a final confirmation that connection is 1 bit = flag active

For Example:

SYN = 1

ACK = 0

that's mean only connection request

TCP assign a number to every packet

like Packet 1 Sequence = 1000, Packet 2 Sequence = 2000, Packet 3 Sequence = 3000

receiver send ACK = 3000

if a packet is missing for example packet 2

then receiver will ACK = 2000 (send packet 2)

UDP :

UDP stands for User Datagram Protocol.

it is connection-less protocol.

In this there is no handshake, no ACK, no re-transmission, no guarantee of packet order.
sender only send the packet

Sender -----> Receiver

That's why UDP is faster than TCP.
it is used in live video call streaming, voice call, Online Games etc

HTTP:

it stands for Hyper Text Transfer protocol.
it is an Application Layer Protocol that exchange data between browser and web server.
When a open a website in browser

Browser -----> Web Server

HTTP Request

Browser <----- Web Server

HTTP Response

HTTP sends data in the plain text.
If an attacker sit between our network then it can see data and also modify it by using MITM.

HTTPS:

HTTPS is also HTTP, but there is SSL and TLS encryption added.
Data will be encrypted before sending

Browser =====Encrypted Data=====> Server

If an attacker capture the packet then it can only see encrypted data

Common Ports:

Port	Protocol	Service	Purpose
20	TCP	FTP (Data)	File transfer (data channel)
21	TCP	FTP (Control)	File transfer (commands/login)
22	TCP	SSH	Secure remote login
23	TCP	Telnet	Remote login (insecure)
25	TCP	SMTP	Sending email
53	TCP/UDP	DNS	Domain name resolution

67	UDP	DHCP Server	Assign IP addresses
68	UDP	DHCP Client	Receive IP address
69	UDP	TFTP	Simple file transfer
80	TCP	HTTP	Web browsing
110	TCP	POP3	Receive email (download)
123	UDP	NTP	Time synchronization
143	TCP	IMAP	Receive and sync email
161	UDP	SNMP	Network monitoring
162	UDP	SNMP Trap	Network alerts
389	TCP/UDP	LDAP	Directory services
443	TCP	HTTPS	Secure web browsing
445	TCP	SMB	Windows file and printer sharing
465	TCP	SMTPS	Secure SMTP
514	UDP	Syslog	Log collection
587	TCP	SMTP Submission	Secure email submission
636	TCP	LDAPS	Secure LDAP
993	TCP	IMAPS	Secure IMAP
995	TCP	POP3S	Secure POP3
1433	TCP	Microsoft SQL Server	Database service
1521	TCP	Oracle Database	Database service
3306	TCP	MySQL	Database service
3389	TCP	RDP	Remote Desktop
5432	TCP	PostgreSQL	Database service
5900	TCP	VNC	Remote desktop
6379	TCP	Redis	In-memory database
8080	TCP	HTTP Alternate	Web applications / Proxy
8443	TCP	HTTPS Alternate	Secure web applications

Stuxnet

Definition

Stuxnet is a highly sophisticated **computer worm (cyber weapon)** discovered in **June 2010**. It was designed to attack **Industrial Control Systems (ICS)** by targeting **Siemens STEP 7 software** and **Programmable Logic Controllers (PLCs)**. It is considered the **first malware known to cause physical damage** to industrial equipment.

Background

Stuxnet targeted **Iran's nuclear enrichment program**. Its goal was to secretly damage uranium centrifuges without launching a military attack.

How Was It Discovered?

- Discovered in **June 2010** by **VirusBlokAda**, a cybersecurity company.
- The malware accidentally spread outside its intended target.
- After analysis by security companies such as **Symantec** and **Kaspersky**, researchers found that it specifically targeted industrial control systems.

How Did Stuxnet Work?

1. Spread through **infected USB drives**.
2. Infected **Windows** computers.
3. Searched for **Siemens STEP 7** software.
4. Modified the program running on **PLCs**.
5. Changed the speed of uranium centrifuges.
6. Displayed **fake normal readings** to operators, hiding the attack.

What is Siemens STEP 7?

Siemens STEP 7 is software used by engineers to **program and configure Siemens PLCs**.

What is a PLC?

A **Programmable Logic Controller (PLC)** is an industrial computer that automatically controls machines in factories, power plants, water treatment plants, and other industrial environments.

Key Features

- First widely known **cyber weapon**
- Targeted **Industrial Control Systems (ICS)**
- Used **multiple zero-day vulnerabilities**
- Spread through **USB drives**
- Used **stolen digital certificates**
- Modified PLC logic

- Caused **physical damage** to industrial equipment

Impact

- Damaged approximately **1,000 uranium centrifuges**
- Delayed Iran's nuclear program
- Demonstrated that cyber attacks can cause **real-world physical damage**
- Increased global focus on **ICS/SCADA security**

Tasks:

Completed Networking Basics from **CISCO Networking Academy**.

